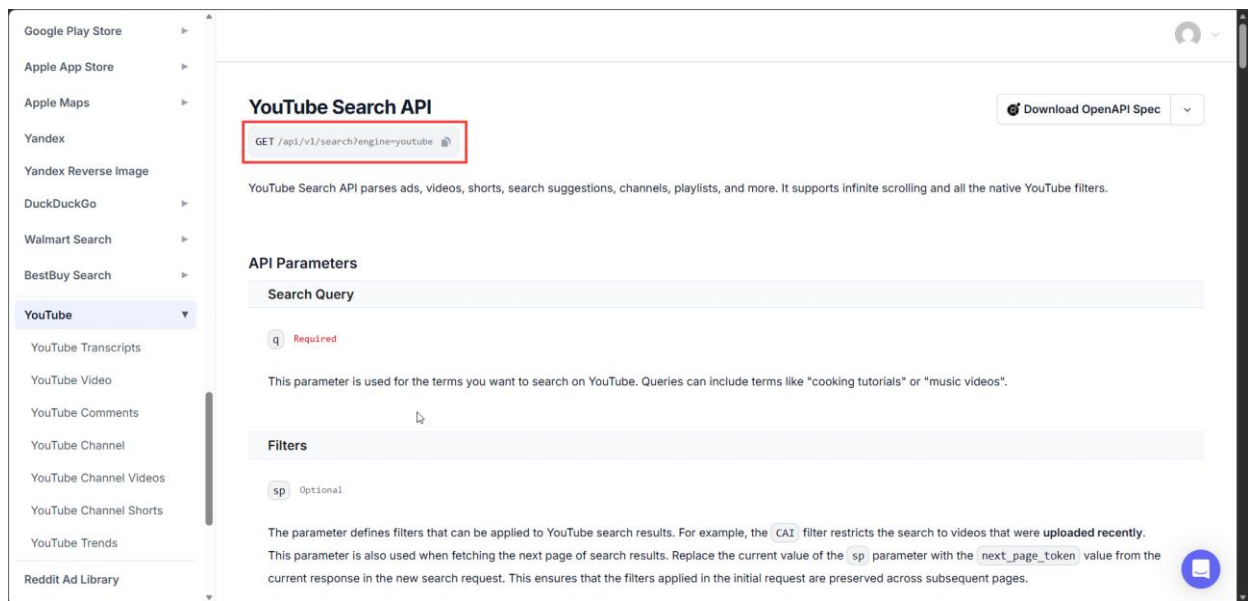


Search API Data Ingestion Pipeline

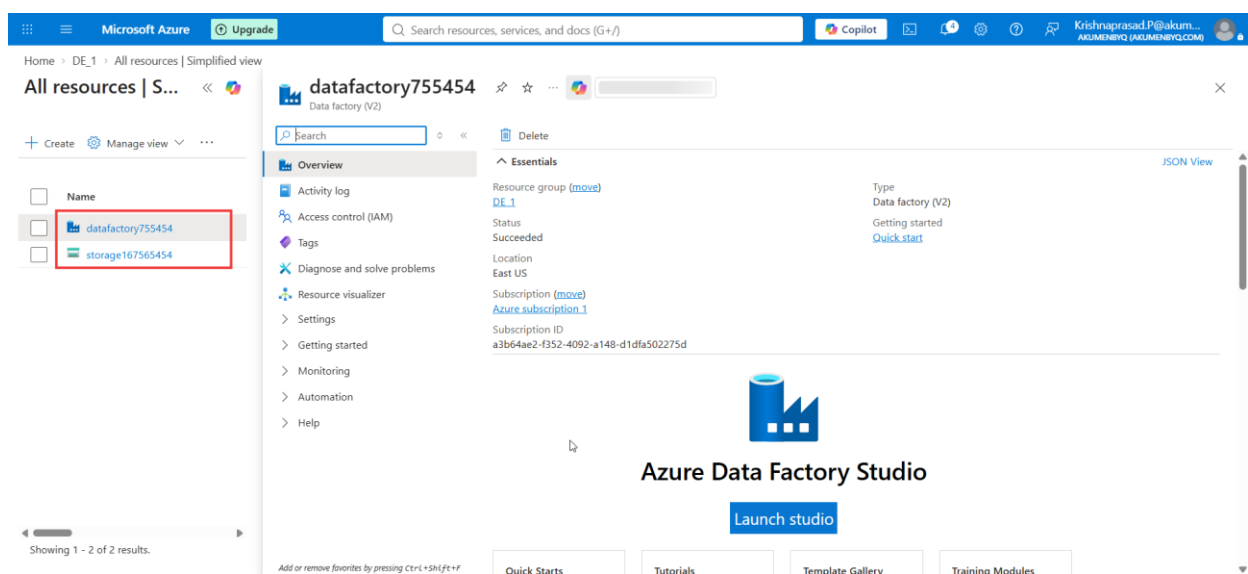
Step 1: Review the Source API

- Reviewed the YouTube Search API on searchapi.io (GET /api/v1/search?engine=youtube), which returns videos, channels, shorts, playlists and more for a given search query.



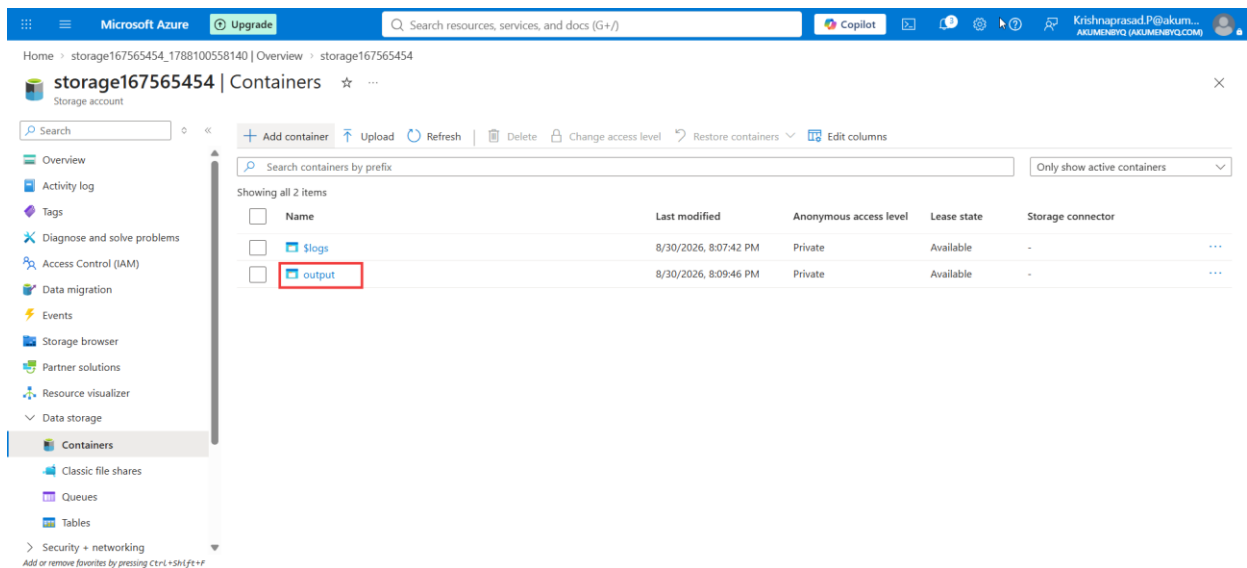
Step 2: Open the Data Factory Resource

- In resource group DE_1, opened datafactory755454 (Data Factory V2) alongside its storage account storage167565454, then launched Azure Data Factory Studio.



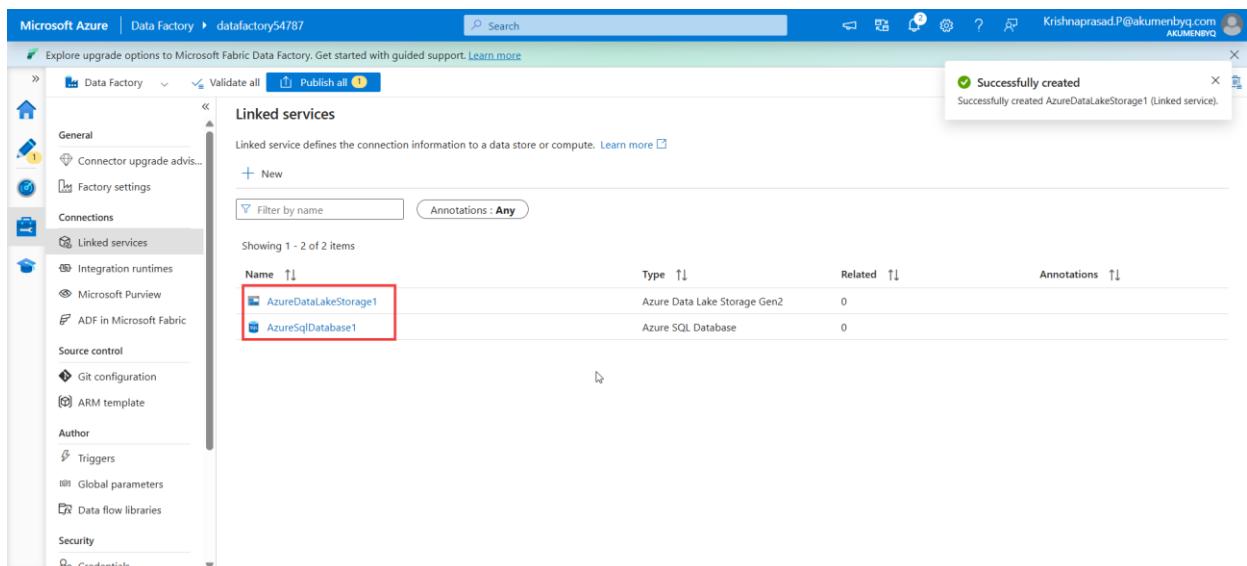
Step 3: Create the Output Container

- In storage account storage167565454, navigated to Containers and created an output container to hold the ingested data, alongside the existing \$logs container.



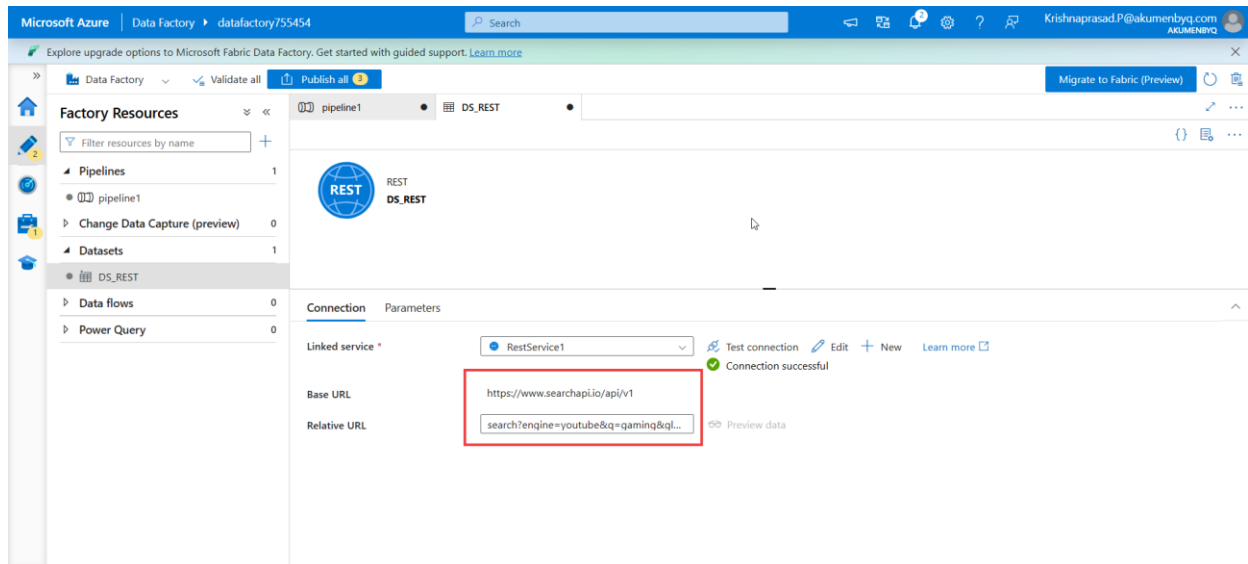
Step 4: Create the Linked Services

- In Data Factory, created two linked services — AzureDataLakeStorage1 (Azure Data Lake Storage Gen2) for the output destination and AzureSqlDatabase1 (Azure SQL Database) — confirming both were created successfully.



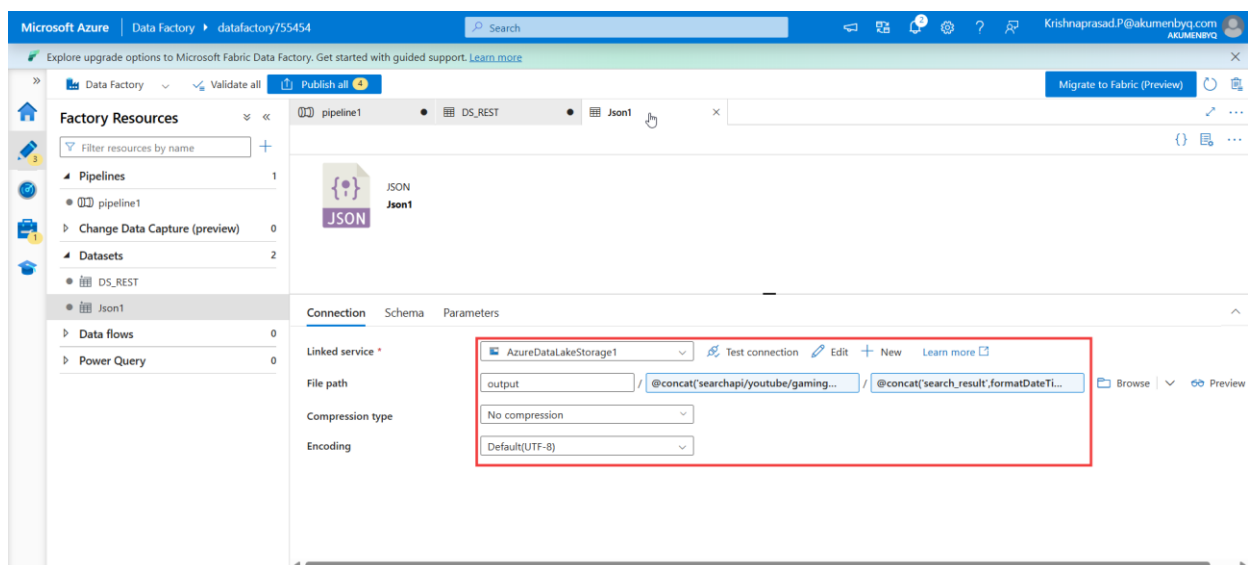
Step 5: Configure the REST Source Dataset

- Created dataset DS_REST using linked service RestService1, with Base URL <https://www.searchapi.io/api/v1> and a Relative URL of `search?engine=youtube&q=gaming&...` to query YouTube for gaming videos, confirming the connection was successful.



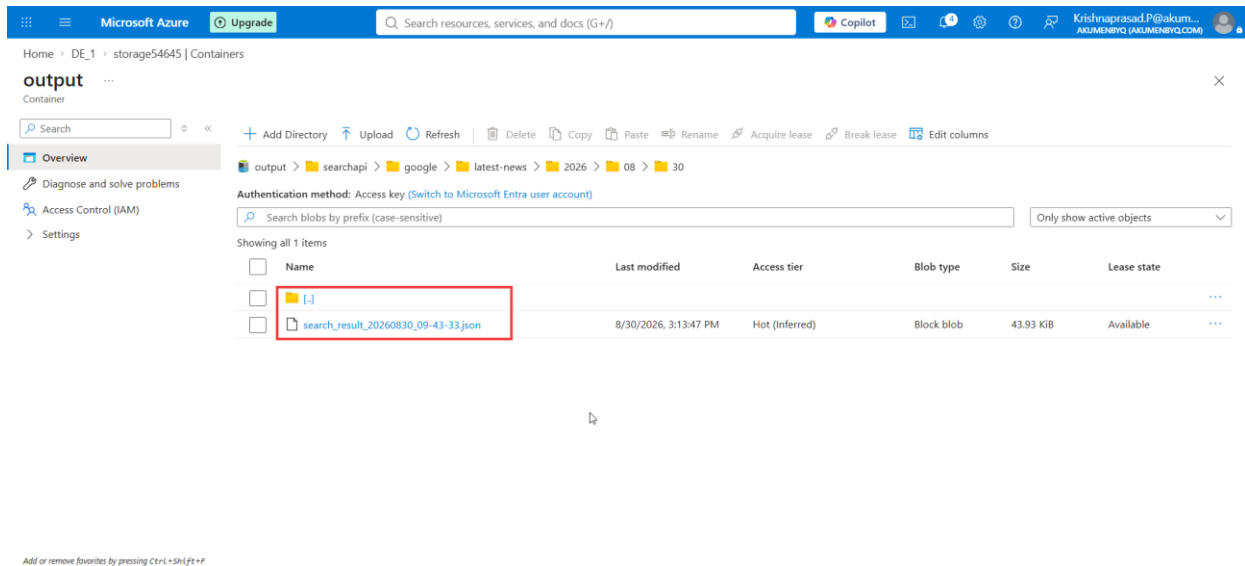
Step 6: Configure the JSON Sink Dataset

- Created dataset Json1 using linked service AzureDataLakeStorage1, with a dynamic file path — `output / @concat('searchapi/youtube/gaming...') / @concat('search_result', formatDateTime(...))` — so each pipeline run is saved as its own timestamped JSON file, organized by engine and query.



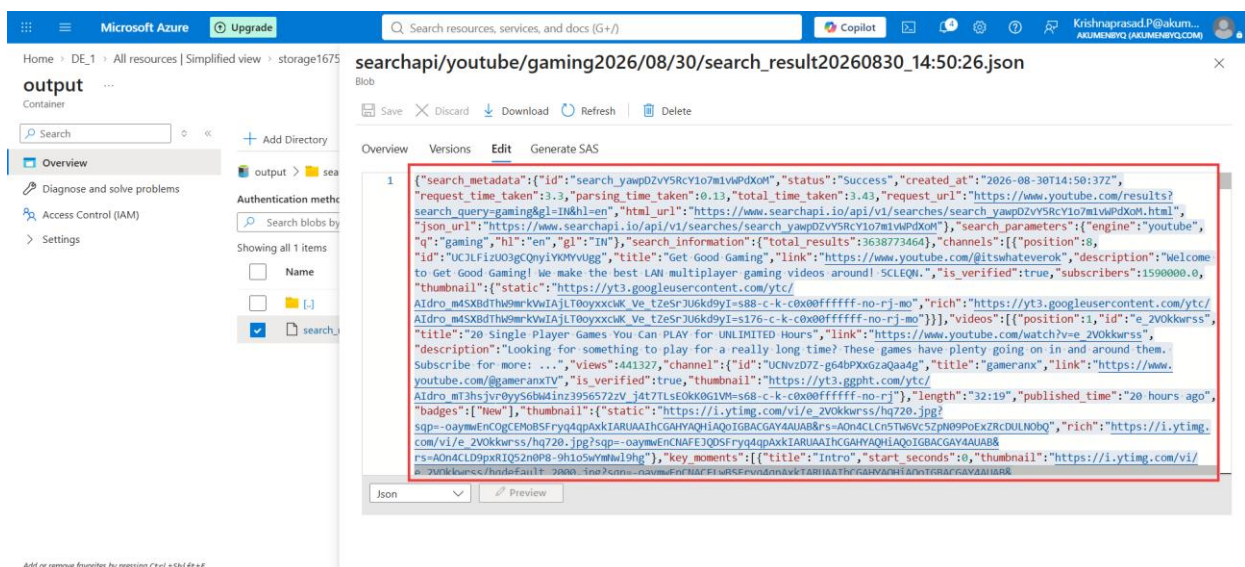
Step 7: Verify the Folder Structure in the Output Container

- Checked the output container and confirmed the pipeline had created the nested folder path searchapi > google > latest-news > 2026 > 08 > 30, containing the resulting file search_result_20260830_09-43-33.json.



Step 8: Inspect the Raw JSON Response

- Opened one of the result files (searchapi/youtube/gaming2026/08/30/search_result20260830_14:50:26.json) to confirm the API response was saved correctly, including search metadata, search parameters, channels and videos.



Step 9: Explore the JSON Schema

- Used the JSON viewer to expand the response structure — search_metadata, search_parameters, search_information, channels and videos — reviewing fields like id, title, link, description, views, channel and thumbnail for each video before mapping.

The screenshot displays a JSON viewer interface. The left pane shows the JSON structure with the following expanded sections:

- `search_metadata`
- `search_parameters`
- `search_information`
- `channels`
- `videos` (array of 2 items)
 - `position: 1`
 - `id`: "e_2VOkKwrs"
 - `title`: "20 Single Player Games You Can PLAY for UNLIMITED Hours"
 - `link`: "https://www.youtube.com/watch?v=e_2VOkKwrs"
 - `description`: "Looking for something to play for a really long time? These games have plenty going on in and around them. Subscribe for more: ..."
 - `views`: 441327
 - `channel`
 - `length`: "32:19"
 - `published_time`: "20 hours ago"
 - `badges`
 - `thumbnail`
 - `key_moments`

The right pane shows a table of the video's metadata:

Name	Value
badges	...
channel	...
description	"Looking for something t..."
id	"e_2VOkKwrs"
key_moments	...
length	"32:19"
link	"https://www.youtube.co..."
position	1
published_time	"20 hours ago"
thumbnail	...
title	"20 Single Player Game..."
views	441327